Xavier Frankline Odhiambo

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Summary

Software engineer with 7 years designing and operating backend systems, from payment integrations and data ingestion pipelines to search infrastructure and LLM-powered services.

Skills

Languages and Frameworks: Python, SQL, JavaScript, Django, Django REST Framework, FastAPI, Flask, Celery
Data: PostgreSQL, ClickHouse, Elasticsearch, Redis, RabbitMQ
Infrastructure: AWS, Docker, Terraform, Ansible, GitHub Actions, GitLab CI, OpenTelemetry

Experience

Senior Software Engineer, Novazury Labs – Nairobi June 2026 – Present

- Built mobile money payment integrations for 3 products with signed webhook verification, idempotent processing, and end-to-end transaction tracing
- Architected a document ingestion pipeline that extracts structured events from unstructured PDFs, routes inconsistent records to human review, and delivers each resulting alert without duplicates on retry
- Standardised background job processing across FastAPI and Django services on PostgreSQL with Procrastinate, removing the need for a separate message broker
- Built CI/CD pipelines for zero-downtime releases that lint, test, and build versioned container images on every change, gate each release on database migrations, and deploy from the image
- Built infrastructure as code with Ansible that provisions every server as an identical, hardened container host with metrics, logs, and traces built in

Senior Software Engineer (Contract), Search Atlas – Remote Aug 2025 – June 2026

- Made scheduled reporting safe under concurrency with Redis-based distributed locks, stopping duplicate customer emails from overlapping Celery Beat runs
- Improved query performance for a customer-facing analytics product through indexing, schema refactors, and migrations on high-volume PostgreSQL and ClickHouse tables
- Secured the analytics REST APIs with validation, authentication, and rate limiting in Django REST Framework
- Added delivery observability to client report emails by tracking Amazon SES delivery, open, click, and bounce events, giving agencies traceability for every report sent
- Improved incident diagnosis with distributed tracing across Django and Celery services using OpenTelemetry, correlated with Sentry, Loki, and Grafana
- Reduced manual status reporting for 2 engineering teams with an LLM-powered tool that turned open issues, errors, and alerts into daily developer briefings and end-of-day reports for the project manager
- Improved the reliability and cost efficiency of LLM features by extending the orchestration layer with automatic provider failover and cost-aware model selection

Senior Software Engineer, Code for Africa – Remote Aug 2022 – Aug 2025

- Built the RabbitMQ and Elasticsearch ingestion service for Mediacloud, an open-source news research pipeline drawing on 34,000 sources, with idempotent URL-hash document IDs that reject duplicate stories
- Architected a multi-stage RAG classification pipeline for ad-domain trust that runs low-cost ranking and link-graph signals first, limiting LlamaIndex and pgvector LLM calls to cases where they change the result
- Led the architecture of an 8-node Elasticsearch cluster (3 master-eligible, 5 data) with Ansible roles for reindexing and lifecycle management, gating cutover on a Rally benchmark of throughput and latency
- Built and tuned the REST APIs behind Sensors.Africa, an IoT air-quality network, and Wazimap-NG, a census mapping platform, removing N+1 queries and adding generated fields on hot read paths

- Led the upgrade of openAFRICA, a CKAN open-data portal, from CKAN 2.8 to 2.11 and Python 2 to 3 as lead maintainer, migrating legacy controllers to Flask blueprints
- Standardised CI/CD across most engineering projects with GitHub Actions and GitLab CI pipelines that lint, test, security-check, and build container images
- Deployed and operated a self-hosted LLM platform (Ollama, OpenWebUI) with role-based access for 5 teams, chosen for data sovereignty and predictable inference costs
- Mentored junior engineers and students through code review and technical guidance

Software Engineer, QED Solutions – Nairobi Apr 2021 – July 2022

- Integrated payment gateways across 8 African markets for a B2B procurement platform, with signed callback verification, idempotent retries, and reconciliation against provider records
- Provisioned AWS infrastructure as code with Terraform and migrated VM workloads to Docker containers on Amazon ECS, so deploys no longer depended on individual servers
- Moved report generation to serverless AWS Lambda functions, taking that load off the main application servers
- Internationalised the platform into English, French, Portuguese, and Amharic with Django internationalisation
- Migrated company websites from WordPress to static hosting on S3 and Lightsail, reducing hosting costs

Software Engineer, Nakala Analytics – Nairobi Aug 2020 – Mar 2021

- Built REST APIs in Flask over PostgreSQL and internal analytics dashboards in Vue.js and React
- Built ETL pipelines with PostgreSQL, dbt, and Stitch, and deployed JupyterHub on Docker Swarm for analysts

Software Engineer, JobsearchKE – Nairobi Sept 2019 – Aug 2020

- Built the company job board, jobsearchke.com, from scratch with Django and PostgreSQL
- Automated email marketing campaigns through the Mailchimp and SendGrid APIs

Projects

Ansible Server Bootstrap github.com/thepsalmist/ansible-server-bootstrap

- Open-sourced the idempotent Ansible roles behind the studio servers, covering SSH hardening with automatic rollback, firewalling, container platform setup, and a metrics, logs, and traces stack

Education

Jomo Kenyatta University of Agriculture and Technology – BS in Telecommunications Engineering Nov 2018

Certifications

AWS Certified Solutions Architect – Associate Jan 2025